Demographic System Modeling in Vensim PLE, Anylogic PLE, and Excel

Dreamland Incorporation
Independent Research Division
"Understanding systems before we change them"

Abstract

The demographic system model is widely used in system dynamics textbooks as an introductory example, yet most sources present only a static diagram without exploring its dynamic behavior or sensitivity to parameter changes. This paper implements the model in three environments — Vensim PLE, AnyLogic PLE, and Excel — to verify its correctness and analyze how variations in birth and death rates affect long-term population dynamics.

The results show that the model exhibits strong sensitivity to the adult death rate, leading to sustained growth or decline depending on parameter values. The analysis also highlights the limitations of the educational version of Vensim (PLE) in handling stock accumulation. The model serves as a foundation for more complex socio-economic systems, such as those linking population dynamics with housing markets, credit availability, and labor conditions.

1. Introduction

The demographic system model – often called the population model – is presented in most system dynamics textbooks as the simplest and most natural starting point for learning about flows and stocks. However, these textbooks rarely say anything other than the general model in Vensim or InsightMaker. Even fewer provide any explanation of this model and examples of how its behavior changes when the parameters are adjusted.

This paper presents the population model in its balanced (reference) state. To verify its correctness, the same model is implemented in three different environments: AnyLogic PLE, Vensim PLE, and Excel. Based on the numerical results obtained in Excel, the influence of key parameters — specifically the birth rate and death rate — is examined.

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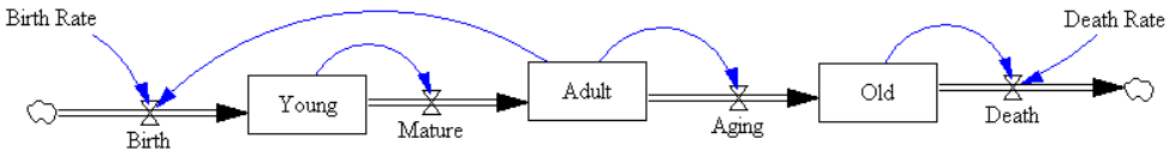
2. Model Description

As mentioned above, textbooks often present the demographic system in the form shown in Figure 1.

Figure 1. Standard demographic system model

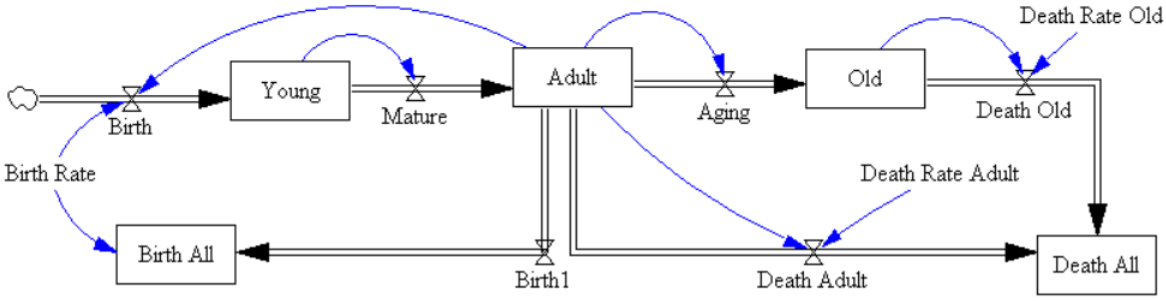


The model consists of three stocks: Young (0-20 years), Adults (20-60 years) and Old (60 years and older); along with flows representing births, mature, aging, and deaths.

There is a single feedback loop in the model, which operates as follows: a larger adult population leads to more births, which increases the young population; after 20 years, these individuals become adults, further increasing the adult population, which in turn leads to even more births. This is a reinforcing loop; it contains no inherent limiting mechanism making the system highly sensitive to changes in the birth and death rates. These two coefficients are the main control parameters of population dynamics in this model.

In the standard model, Death is applied only to the old population. This is corrected here by adding Death among the adult population, as the most numerous group. For further calculations and analysis of the values, Death and Birth are considered as stocks that show the total number of births and deaths during the period. The Birth will be considered from the adult stock. The final structure of the working model is shown in Figure 2.

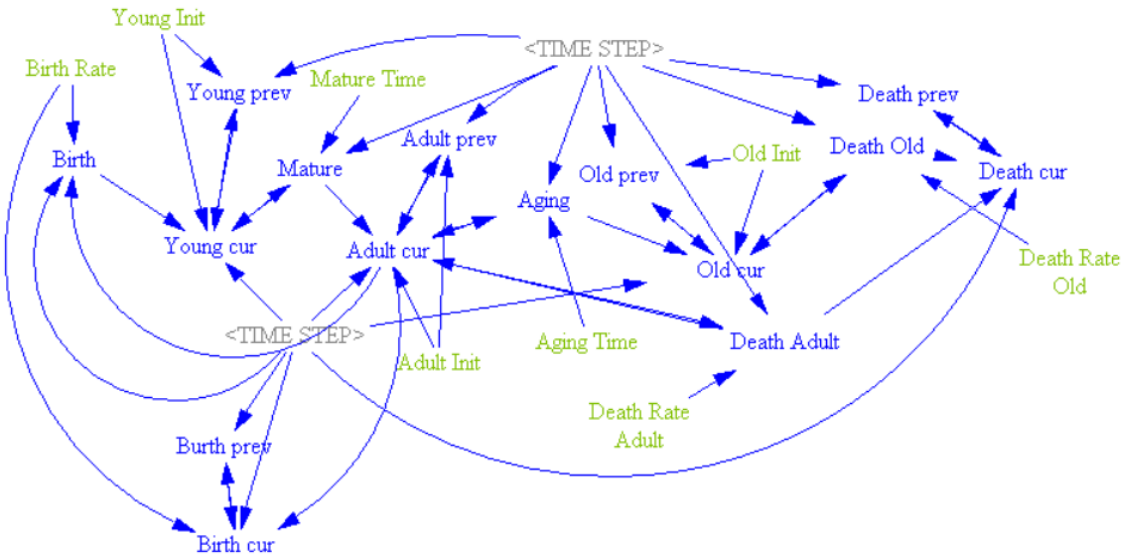
Figure 2. Working model of the demographic system



Vensim PLE virsion does not support the built-in INTEG function for stock accumulating. To work around this, additional variables were introduced to emulate integration using the DELAY FIXED function. This approach yields numerical results equivalent to continuous integration, but requires explicit

separation of current and previous stock values. The computational scheme is shown in Figure 3. Green-colored elements indicate the initial conditions (Table 1), which are consistent across all model implementations.

Figure 3. Computational model of the demographic system in Vensim PLE



As illustrated Figure 3, each stock (Young, Adult, Old, Birth All, Death All) is split into the previous and current values, ensuring that the accumulated value correctly updates at each time step.

A similar model was built in AnyLogic PLE to verify the results. It follows the same logic but uses a built-in integrator, eliminating the need for additional tracking variables.

Table 1. Initial values

Parameter	Value	Units
Time Step	0.25	Year
Initial Population	1350	People
Young Initial	300	People
Adult Initial	1000	People
Old Initial	50	People
Birth Rate	0.035	
Death Rate Old	0.01	
Death Rate Adult	0.01	
Mature Time	20	Year
Aging Time	40	Year
Final Time	100	Year

3. Results

3.1. Vensim PLE

Due to the limitations of Vensim PLE (the INTEG function is not available) errors accumulate over time, and the results exhibit doubling at every second time step. This leads to distorted graphs and numerical values, as shown in Figures 4 and 5.

Figure 4. Numerical values obtained in Vensim PLE

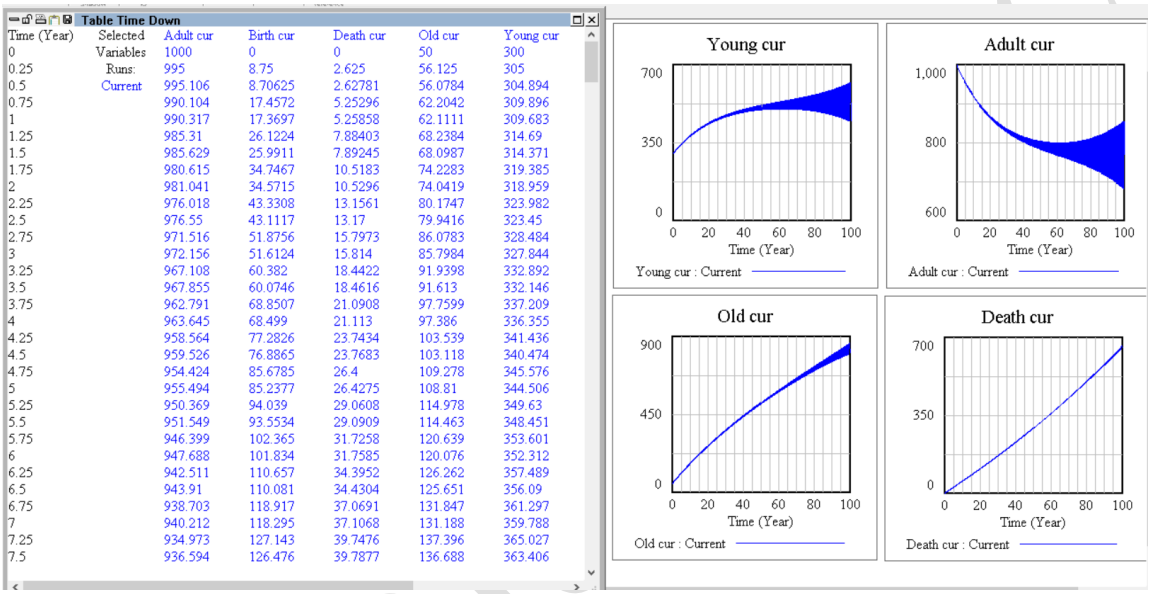
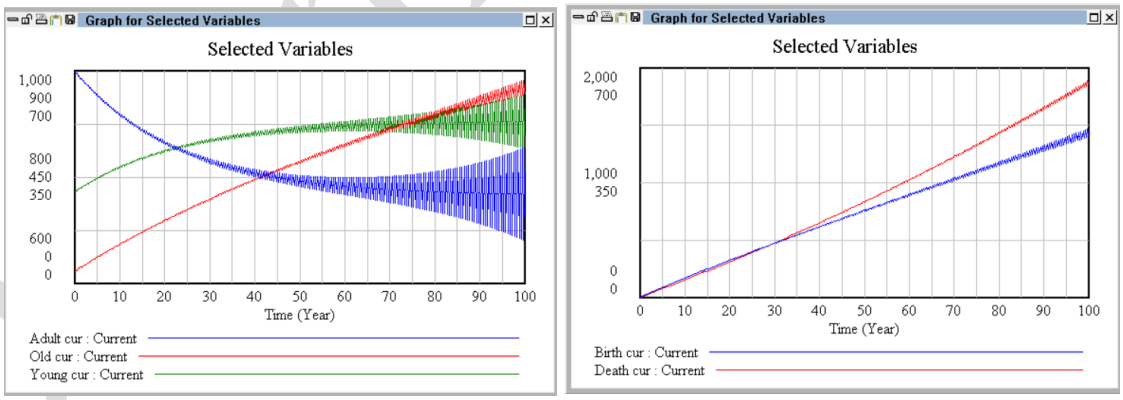


Table Time Down

Time (Year)	Selected Variables	Adult cur	Birth cur	Death cur	Old cur	Young cur
0		1000	0	0	50	300
0.25	Runs:	995	8.75	2.625	56.125	305
0.5	Current	995.106	8.70625	2.62781	56.0784	304.894
0.75		990.104	17.4572	5.25296	62.2042	309.896
1		990.317	17.3697	5.25858	62.1111	309.683
1.25		985.31	26.1224	7.88403	68.2384	314.69
1.5		985.629	25.9911	7.89245	68.0987	314.371
1.75		980.615	34.7467	10.5183	74.2283	319.385
2		981.041	34.5715	10.5296	74.0419	318.959
2.25		976.018	43.3308	13.1561	80.1747	323.982
2.5		976.55	43.1117	13.17	79.9416	323.45
2.75		971.516	51.8756	15.7973	86.0783	328.484
3		972.156	51.6124	15.814	85.7984	327.844
3.25		967.108	60.382	18.4422	91.9398	332.892
3.5		967.855	60.0746	18.4616	91.613	332.146
3.75		962.791	68.8507	21.0908	97.7599	337.209
4		963.645	68.499	21.113	97.386	336.355
4.25		958.564	77.2826	23.7434	103.539	341.436
4.5		959.526	76.8865	23.7683	103.118	340.474
4.75		954.424	85.6785	26.4	109.278	345.576
5		955.494	85.2377	26.4275	108.81	344.506
5.25		950.369	94.039	29.0608	114.978	349.63
5.5		951.549	93.5534	29.0909	114.463	348.451
5.75		946.399	102.365	31.7258	120.639	353.601
6		947.688	101.834	31.7585	120.076	352.312
6.25		942.511	110.657	34.3952	126.262	357.489
6.5		943.91	110.081	34.4304	125.651	356.09
6.75		938.703	118.917	37.0691	131.847	361.297
7		940.212	118.295	37.1068	131.188	359.788
7.25		934.973	127.143	39.7476	137.396	365.027
7.5		936.594	126.476	39.7877	136.688	363.406

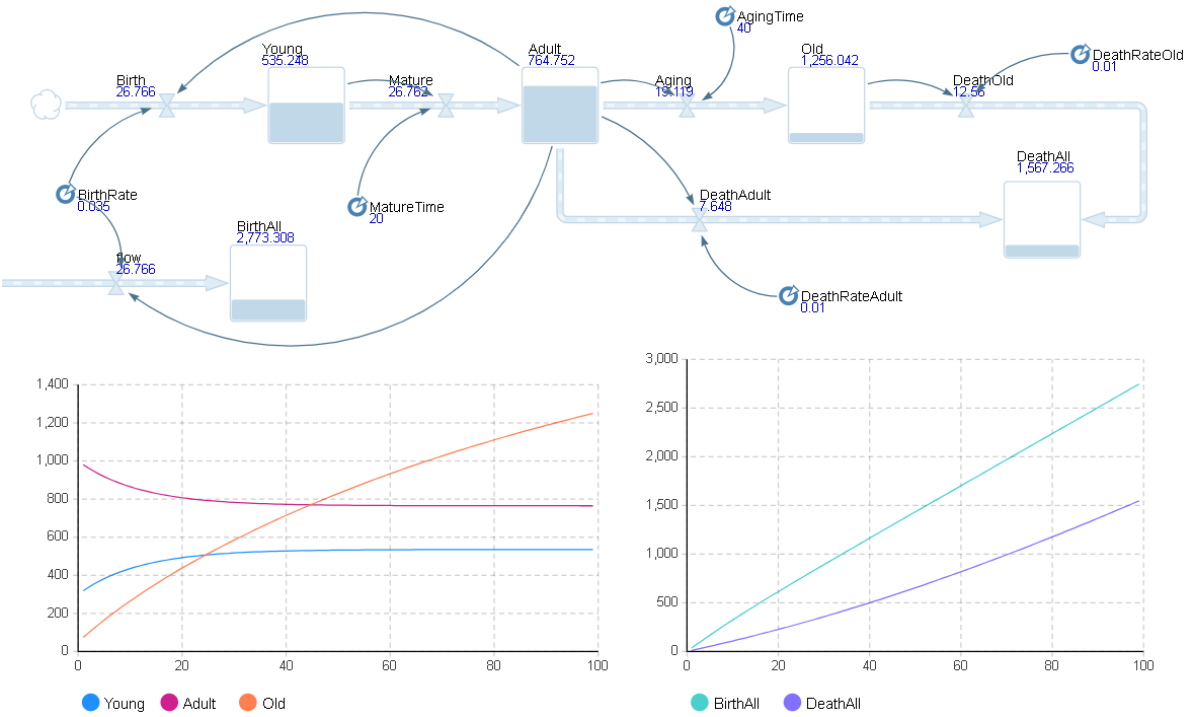
Figure 5. Stock graphs obtained in Vensim PLE



3.2. AnyLogic PLE

The AnyLogic model shown in Figure 6 uses a built-in integrator, eliminating the need for additional tracking variables and producing correct graphs.

Figure 6. Demographic system model and its graphs in AnyLogic PLE



3.3. Excel

The same initial values were used for the Excel verification. All stocks (Young, Adult, Old, Birth All, Death All) were calculated for the first step (Time Step = 0.25) using the following formulas:

$Young_{cur} = Young_{prev} + Birth_{cur} – Mature_{cur} = 300 + 8.75 – 3.75 = 305$

$Adult_{cur} = Adult_{prev} + Mature_{cur} – Aging_{cur} – DeathAdult_{cur} =$

$= 1000 + 3.75 – 6.25 - 2.5 = 995$

$Old_{cur} = Old_{prev} + Aging_{cur} – DeathOld_{cur} = 50 + 6.25 - 0.13 = 56.13$

$DeathAll_{cur} = DeathAll_{prev} + DeathOld_{cur} + DeathAdult_{cur} = 0+0.13+2.5 = 2.63$

$BirthAll_{cur} = BirthAll_{prev} + Birth_{cur} = 0 + 8.75 = 8.75$

To verify the calculations, two indicators were used: Expected Population and Total Population, which were calculated using the following formulas:

Expected Population = Initial Population + BirthAll– DeathAll

Total Population = Young + Adult + Old

A zero Gap between these two indicators confirms the correctness of the model and the calculations. For the first step:

$Expected\ Population_{cur} = Initial\ Population_{cur} + BirthAll_{cur} – DeathAll_{cur} =$

$= 1350 + 8.75 – 2.63 = 1356.13$

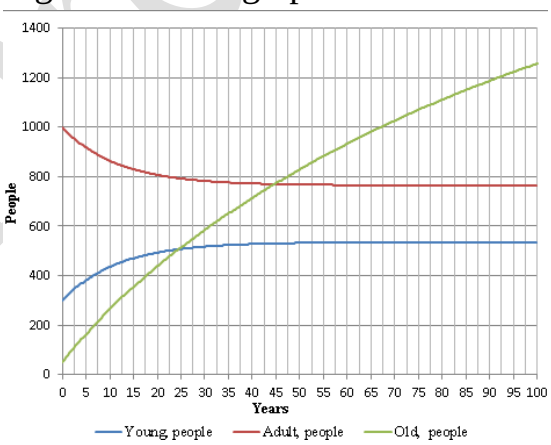
Total $\text{Population}_{cur} = \text{Young}_{cur} + \text{Adult}_{cur} + \text{Old}_{cur} =$
$= 305 + 995 + 56.13 = 1356.13$
Gap = Expected Population_{cur} – Total Population_{cur} = 1356.13 – 1356.13 = 0

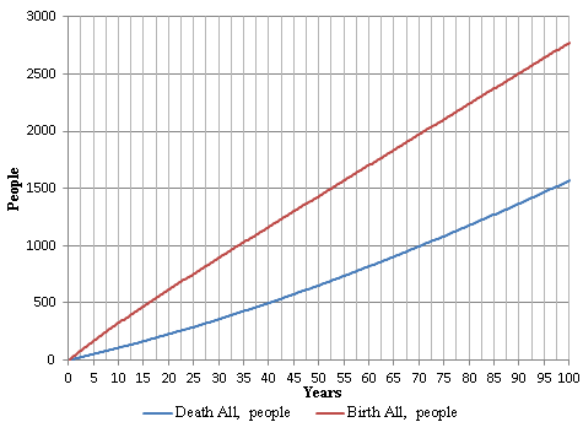
Numerical values for the first 36 steps are shown in Figure 7, and the corresponding graphs (over a 100-year horizon) in Figure 8.

Figure 7. Numerical values obtained in Excel

					Birth	Young, people	Mature	Adult, people	Aging	Old, people	Death_ Old, people	Death_ Adult, people	Death All, people	Birth All, people	Expected population, people	Total population, people	Gap
Time Step	0,25	Year	0	0	0	300	0	1 000,00	0,00	50,00	0,00	0,00	0,00	0,00	1 350,00	1 350,00	0,00
Init.Population	1350	People	1	0,25	8,75	305,00	3,75	995,00	6,25	56,13	0,13	2,50	2,63	8,75	1356,13	1356,13	0,00
Young Init	300	People	2	0,50	8,71	309,89	3,81	990,11	6,22	62,20	0,14	2,49	5,25	17,46	1362,20	1362,20	0,00
Adult Init	1000	People	3	0,75	8,66	314,68	3,87	985,32	6,19	68,24	0,16	2,48	7,88	26,12	1368,24	1368,24	0,00
Old Init	50	People	4	1,00	8,62	319,37	3,93	980,63	6,16	74,22	0,17	2,46	10,52	34,74	1374,22	1374,22	0,00
Birth Rate	0,035		5	1,25	8,58	323,96	3,99	976,04	6,13	80,17	0,19	2,45	13,15	43,32	1380,17	1380,17	0,00
Death Rate Old	0,01		6	1,50	8,54	328,45	4,05	971,55	6,10	86,07	0,20	2,44	15,80	51,86	1386,07	1386,07	0,00
Death Rate Adult	0,01		7	1,75	8,50	332,85	4,11	967,15	6,07	91,92	0,22	2,43	18,44	60,36	1391,92	1391,92	0,00
Mature time	20	Year	8	2,00	8,46	337,15	4,16	962,85	6,04	97,74	0,23	2,42	21,09	68,83	1397,74	1397,74	0,00
Aging time	40	Year	9	2,25	8,42	341,36	4,21	958,64	6,02	103,51	0,24	2,41	23,74	77,25	1403,51	1403,51	0,00
			10	2,50	8,39	345,48	4,27	954,52	5,99	109,25	0,26	2,40	26,39	85,64	1409,25	1409,25	0,00
			11	2,75	8,35	349,51	4,32	950,49	5,97	114,94	0,27	2,39	29,05	93,99	1414,94	1414,94	0,00
			12	3,00	8,32	353,46	4,37	946,54	5,94	120,59	0,29	2,38	31,72	102,31	1420,59	1420,59	0,00
			13	3,25	8,28	357,33	4,42	942,67	5,92	126,21	0,30	2,37	34,38	110,59	1426,21	1426,21	0,00
			14	3,50	8,25	361,11	4,47	938,89	5,89	131,78	0,32	2,36	37,06	118,84	1431,78	1431,78	0,00
			15	3,75	8,22	364,81	4,51	935,19	5,87	137,32	0,33	2,35	39,73	127,05	1437,32	1437,32	0,00
			16	4,00	8,18	368,43	4,56	931,57	5,84	142,82	0,34	2,34	42,41	135,24	1442,82	1442,82	0,00
			17	4,25	8,15	371,98	4,61	928,02	5,82	148,29	0,36	2,33	45,10	143,39	1448,29	1448,29	0,00
			18	4,50	8,12	375,45	4,65	924,55	5,80	153,72	0,37	2,32	47,79	151,51	1453,72	1453,72	0,00
			19	4,75	8,09	378,84	4,69	921,16	5,78	159,11	0,38	2,31	50,49	159,60	1459,11	1459,11	0,00
			20	5,00	8,06	382,17	4,74	917,83	5,76	164,47	0,40	2,30	53,19	167,66	1464,47	1464,47	0,00
			21	5,25	8,03	385,42	4,78	914,58	5,74	169,80	0,41	2,29	55,89	175,69	1469,80	1469,80	0,00
			22	5,50	8,00	388,61	4,82	911,39	5,72	175,09	0,42	2,29	58,60	183,69	1475,09	1475,09	0,00
			23	5,75	7,97	391,72	4,86	908,28	5,70	180,35	0,44	2,28	61,32	191,67	1480,35	1480,35	0,00
			24	6,00	7,95	394,78	4,90	905,22	5,68	185,57	0,45	2,27	64,04	199,61	1485,57	1485,57	0,00
			25	6,25	7,92	397,76	4,93	902,24	5,66	190,76	0,46	2,26	66,77	207,53	1490,76	1490,76	0,00
			26	6,50	7,89	400,68	4,97	899,32	5,64	195,93	0,48	2,26	69,50	215,43	1495,93	1495,93	0,00
			27	6,75	7,87	403,54	5,01	896,46	5,62	201,06	0,49	2,25	72,24	223,30	1501,06	1501,06	0,00
			28	7,00	7,84	406,34	5,04	893,66	5,60	206,16	0,50	2,24	74,98	231,14	1506,16	1506,16	0,00
			29	7,25	7,82	409,08	5,08	890,92	5,59	211,23	0,52	2,23	77,73	238,96	1511,23	1511,23	0,00
			30	7,50	7,80	411,77	5,11	888,23	5,57	216,27	0,53	2,23	80,49	246,76	1516,27	1516,27	0,00
			31	7,75	7,77	414,39	5,15	885,61	5,55	221,28	0,54	2,22	83,25	254,53	1521,28	1521,28	0,00
			32	8,00	7,75	416,96	5,18	883,04	5,54	226,26	0,55	2,21	86,02	262,28	1526,26	1526,26	0,00
			33	8,25	7,73	419,48	5,21	880,52	5,52	231,21	0,57	2,21	88,79	270,00	1531,21	1531,21	0,00
			34	8,50	7,70	421,94	5,24	878,06	5,50	236,14	0,58	2,20	91,57	277,71	1536,14	1536,14	0,00
			35	8,75	7,68	424,35	5,27	875,65	5,49	241,04	0,59	2,20	94,36	285,39	1541,04	1541,04	0,00
			36	9,00	7,66	426,70	5,30	873,30	5,47	245,91	0,60	2,19	97,15	293,05	1545,91	1545,91	0,00

Figure 8. Stock graphs obtained in Excel





As can be seen from Figures 6 and 8, the graphs obtained in Excel match the graphs in AnyLogic, which means that the calculations are correct.

4. Parameter Variation

Using the Excel model, we now examine how changes in the birth rate, old-age death rate, and adult death rate affect population dynamics.

Increasing the Birth Rate from 0.035 to 0.037 over a 100-year horizon (Figure 9) produces only a modest population increase. However, when the simulation is extended to 300 years (Figure 10), the reinforcing feedback loop becomes more pronounced, gradually driving system growth.

Figure 9. Stock graphs for Birth Rate = 0.037, 100-year horizon

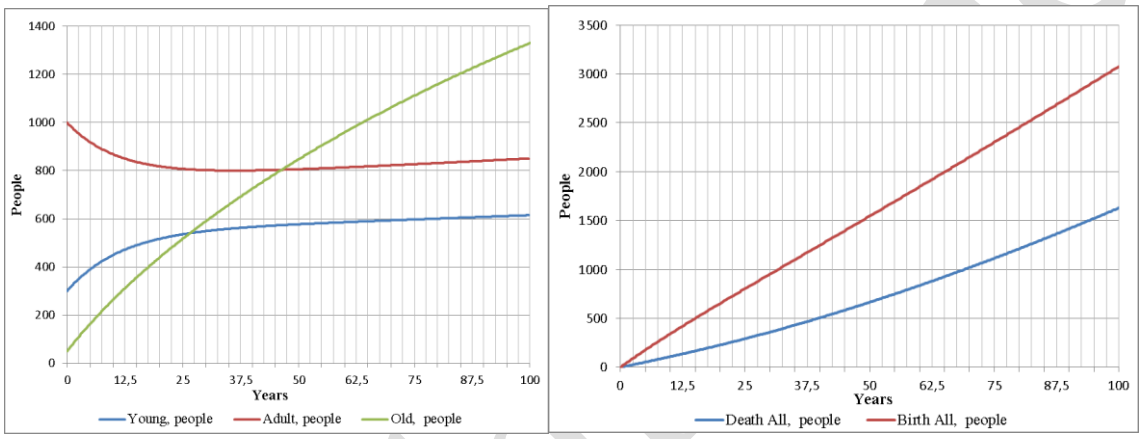
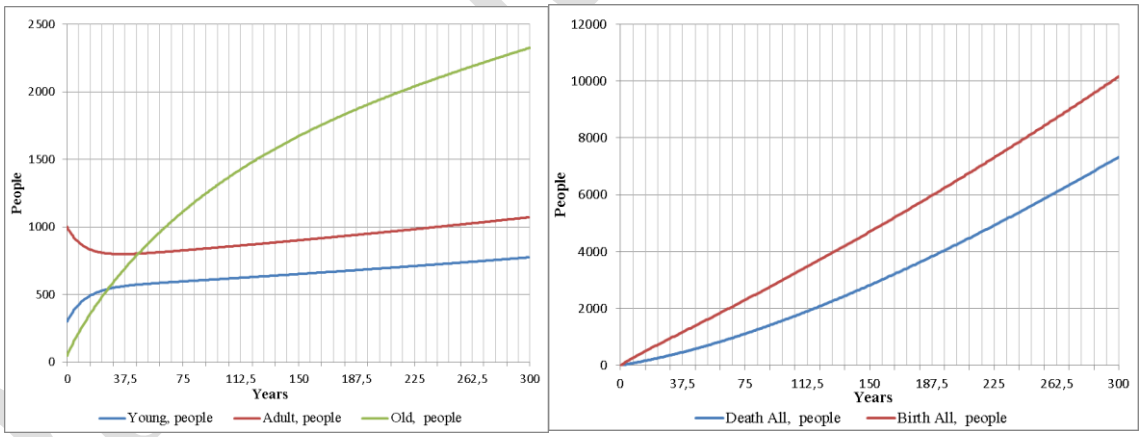
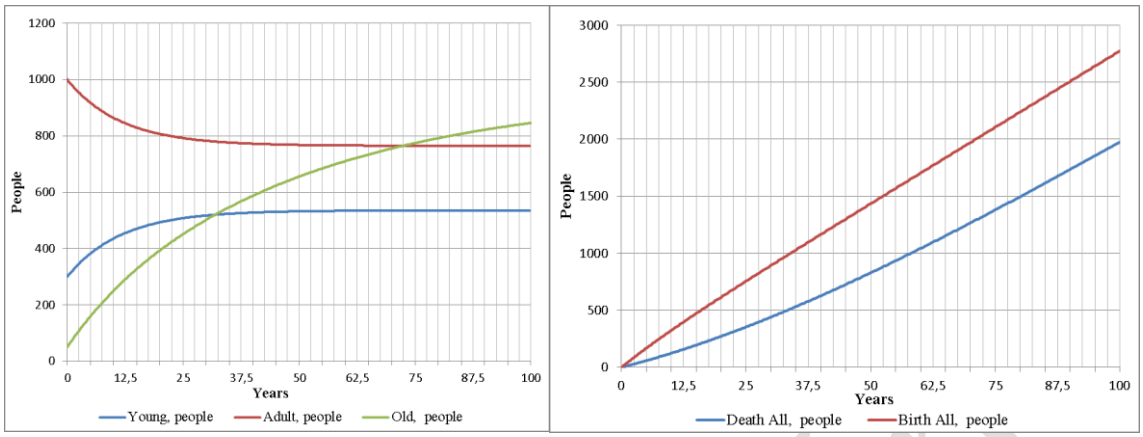


Figure 10. Stock graphs for Birth Rate = 0.037, 300-year horizon



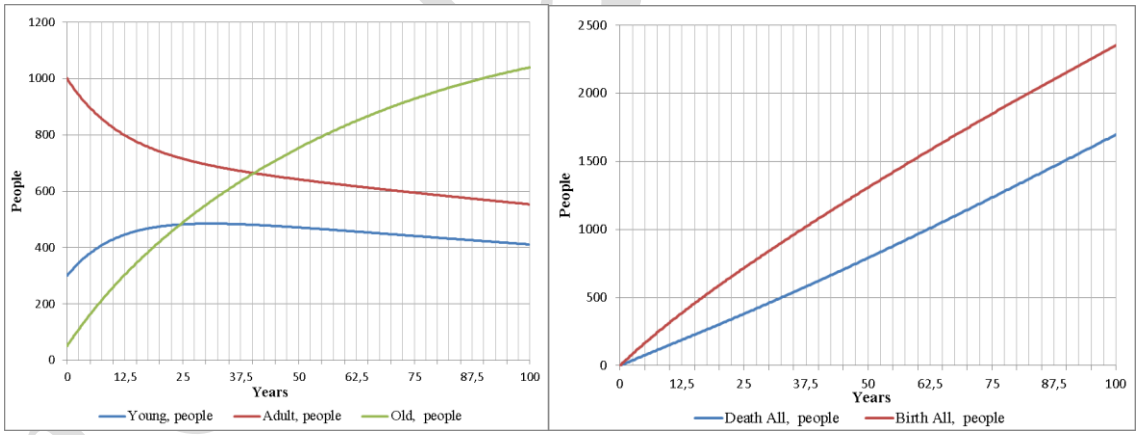
An increase in the Old Death Rate from 0.01 to 0.02 affects only the Death Old and Death All indicators, without affecting other stocks, as shown in Figure 11. This is expected behavior; the Old stock has no direct influence on Birth in the presented model.

Figure 11. Stock graphs for Death Rate Old = 0.02



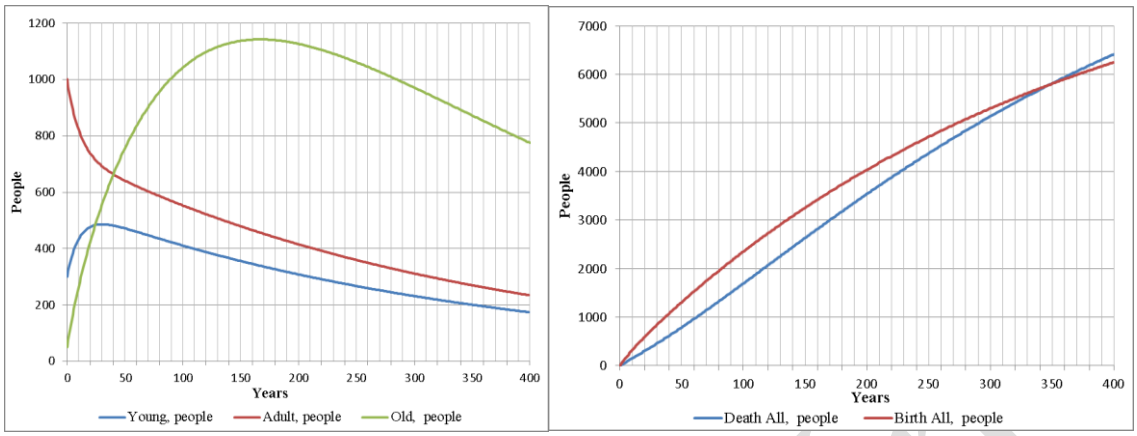
The system will behave differently with an increase in Death Rate Adult. If the Adult Death Rate is increased from 0.01 to 0.015 (Figure 12), both the Birth (through the reduction in adult population) and the Death All are affected. A feedback loop is visible on the Birth flow, where fewer adults produce fewer children, which further reduces the number of adults even more, and, consequently, there are even fewer children.

Figure 12. Stock graphs for Death Rate Adult = 0.015



The long-term consequences become most visible when the simulation horizon is extended to 400 years (Figure 13). The system enters a sustained decline when the Death All stock begins to exceed the Birth All stock.

Figure 13. Stock graphs for Death Rate Adult = 0.015, 400-year horizon



5. Conclusion

The demographic system model – often called the population model – is considered a cornerstone of system dynamics for good reason. By adjusting the birth and death rates, long-term population trends can be projected over decades, assuming external conditions remain unchanged.

This simple structure serves as the foundation for more complex models. For instance, birth rates can be linked to housing affordability, which in turn depends on access to credit, wage levels, working hours, and interest rates. Such connections create feedback loops where, for example, the constant employment of the adult population may ultimately hinder reproduction.

Yet no matter how complex these extensions become, the core remains the simple population model presented in this work.

Acknowledgments

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